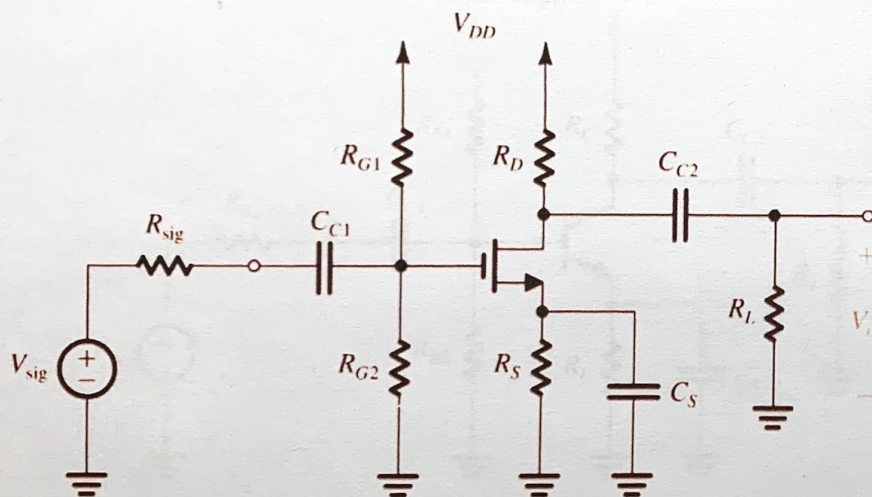


作业

$$R_G = R_{G1} \parallel R_{G2}$$

10.26 A CS amplifier has $C_{C1} = C_S = C_{C2} = 1 \mu\text{F}$, $R_G = 10 \text{ M}\Omega$, $R_{\text{sig}} = 100 \text{ k}\Omega$, $g_m = 2 \text{ mA/V}$, $R_D = R_L = R_S = 10 \text{ k}\Omega$. Find A_M , f_{P1} , f_{P2} , f_{P3} , f_Z , and f_L .

Ans. -9.9 V/V ; 0.016 Hz ; 334.2 Hz ; 8 Hz ; 15.91 Hz ; 334.2 Hz



(a)

$$\frac{V_G}{V_{\text{sig}}} = \frac{10\text{M}}{10\text{M} + 100\text{k}} = 0.99$$

$$A_M = \frac{V_G}{V_{\text{sig}}} \cdot [-g_m(R_D \parallel R_L)] = -9.9 \text{ V/V}$$

$$f_{P1} = \frac{1}{2\pi C_{C1}(R_{G1} + R_{\text{sig}})} = 0.016 \text{ Hz}$$

$$f_{P2} = \frac{1}{2\pi C_S \cdot (R_S \parallel \frac{1}{g_m})} = 334.4 \text{ Hz}$$

$$f_{P3} = \frac{1}{2\pi C_{C2} \cdot (R_D + R_L)} = 7.96 \text{ Hz}$$

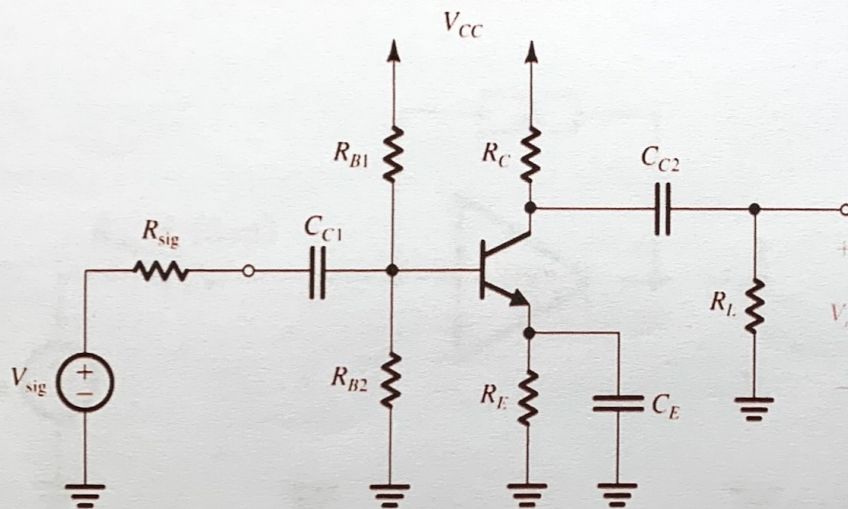
$$f_Z = \frac{1}{2\pi C_S R_S} = 15.92 \text{ Hz}$$

$$\therefore f_{P2} \gg f_{P3}, f_{P1} \quad \therefore f_L = f_{P2} = 334.4 \text{ Hz}$$

$$R_B = R_{B1} || R_{B2}$$

10.27 A common-emitter amplifier has $C_{C1} = C_E = C_{C2} = 1 \mu\text{F}$, $R_B = 100 \text{ k}\Omega$, $R_{\text{sig}} = 5 \text{ k}\Omega$, $g_m = 40 \text{ mA/V}$, $r_\pi = 2.5 \text{ k}\Omega$, $R_E = 5 \text{ k}\Omega$, $R_C = 8 \text{ k}\Omega$, and $R_L = 5 \text{ k}\Omega$. Find the value of the time constant associated with each capacitor, and hence estimate the value of f_L . Also compute the frequency of the transmission zero introduced by C_E and comment on its effect on f_L .

Ans. $\tau_{C1} = 7.44 \text{ ms}$; $\tau_{CE} = 0.071 \text{ ms}$; $\tau_{C2} = 13 \text{ ms}$; $f_L = 2.28 \text{ kHz}$; $f_Z = 31.8 \text{ Hz}$, which is much smaller than f_L and thus has a negligible effect on f_L .



$$R_{C1} = R_B || r_\pi + R_{\text{sig}} = 100 \text{ k} / / 2.5 \text{ k} + 5 \text{ k} = 7.439 \text{ k}$$

$$\tau_{C1} = R_{C1} \cdot C_{C1} = 7.439 \text{ ms}$$

$$\beta = r_\pi \cdot g_m = 2.5 \text{ k} \cdot 40 \text{ m} = 100$$

$$R_{CE} = R_E || \left[\frac{R_B || R_{\text{sig}}}{\beta + 1} + \frac{1}{g_m} \right] = 71.122 \Omega$$

$$\tau_{CE} = R_{CE} \cdot C_{CE} = 0.071 \text{ ms}$$

$$R_{C2} = R_C + R_L = 13 \text{ k}$$

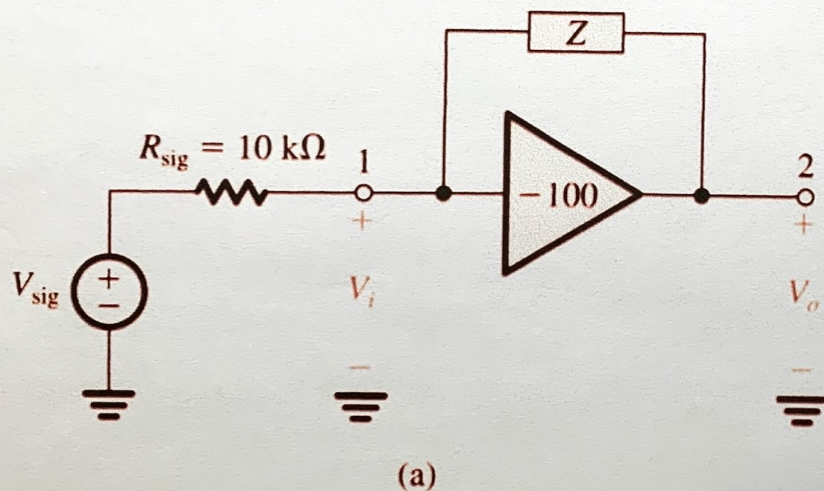
$$\tau_{C2} = R_{C2} \cdot C_{C2} = 13 \text{ ms}$$

$$f_L = \frac{1}{2\pi} \left[\frac{1}{\tau_{C1}} + \frac{1}{\tau_{CE}} + \frac{1}{\tau_{C2}} \right] = 2.276 \text{ kHz}$$

$$f_Z = \frac{1}{2\pi C_E R_E} = 31.847 \text{ Hz}$$

Example 10.4

Figure 10.15(a) shows an ideal voltage amplifier with a gain of -100 V/V and an impedance Z connected between its output and input terminals. Find the Miller equivalent circuit when Z is (a) a $1\text{-M}\Omega$ resistance and (b) a 1-pF capacitance. In each case, use the equivalent circuit to determine V_o/V_{sig} .



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